

Juan Cruz Ferreyra

857-376-1180 | ferreyra.juancruz95@gmail.com | [linkedin.com/in/jcruz-ferreyra/](https://www.linkedin.com/in/jcruz-ferreyra/) | github.com/jcruz-ferreyra

WORK EXPERIENCE

Research Assistant <i>Boston Area Research Initiative</i> - Owned end-to-end development and deployment of real-time data pipeline and interactive dashboard for 50+ distributed environmental sensors. Designed cloud infrastructure for data ingestion and transformation, implemented relational database with normalized schema and automated data quality validation, and built web-based dashboard enabling researchers and community stakeholders to independently query and visualize environmental data. - Led product rollout including comprehensive documentation (data dictionaries, operational procedures, system specifications), user training for researchers, and system handoff. Monitor and maintain pipeline to ensure data integrity and reliable performance for ongoing environmental monitoring. - Curated and published open datasets by integrating heterogeneous city data sources with BARI's geospatial infrastructure linking parcel boundaries to census geographies (blocks, block groups, tracts). Performed spatial joins and data standardization across tabular administrative records and geospatial layers, publishing datasets to research dataverse with comprehensive documentation. <i>Python, R, SQL, Azure, Streamlit, ArcGIS</i>	Jan. 2025 – Present <i>Boston, Massachusetts</i>
Data Scientist <i>Rosario Mobility Authority</i> - Analyzed GPS trajectory data, smart card transactions, and operational logs from 400+ buses across relational database systems to monitor transit system performance and identify service disruptions. Collaborated with operations and planning teams to build automated dashboards and statistical reports used by executive leadership for data-driven decision-making. - Built predictive model combining spatial and temporal features with gradient boosting to recover missing origin locations in large-scale GPS datasets (2M+ records). Performed feature engineering achieving 93.8% accuracy, improving data completeness by 15%+ and enabling more precise origin-destination analysis for underrepresented geographical areas. - Designed and implemented automated data quality system to detect and correct GPS trajectory errors, developing spatial route-based heuristics. Reduced manual data cleaning effort by 80%, ensuring database integrity and improving data reliability for operational analytics. - Conducted statistical analysis on 4+ years of bike-sharing trip data to identify complex usage patterns accounting for 12.6% of vehicle losses. Built user segmentation model achieving 99%+ F1-score, documented findings, and presented data-driven recommendations to stakeholders through technical reports. <i>Python (Pandas, GeoPandas, Scikit-learn), R, SQL, PostgreSQL, Git</i>	Nov. 2022 – Aug. 2024 <i>Rosario, Argentina</i>

EDUCATION

Northeastern University <i>Master of Science, Urban Informatics — Fulbright Scholar (GPA: 4.0)</i>	Boston, MA <i>Expected May 2026</i>
University of Buenos Aires <i>Graduate Specialization, Artificial Intelligence</i>	Buenos Aires, Argentina <i>October 2025</i>
National University of Rosario <i>Bachelor in Architecture</i>	Rosario, Argentina <i>May 2021</i>

PROJECTS

Computer Vision for Urban Mobility Classification Developed detection and classification system for vehicle counting and cyclist demographic inference using machine learning, in collaboration with public transit agency. Performed iterative model evaluation and feature engineering, achieving 2x performance improvement in relevant classes while reducing manual labeling effort by 60% through active learning pipeline. <i>Python (PyTorch, OpenCV, Scikit-learn), Jupyter</i>	Jan. 2025 – Oct. 2025
Open Source Contribution: Supervision Library Contributed feature enhancements to production computer vision library with 35K+ GitHub stars. Refactored core component to support flexible detection countings through code review process with maintainers, demonstrating strong Python programming and collaborative development skills (+300/-80 LOC). <i>Python (OpenCV, NumPy), Git</i>	Feb. 2024 – Aug. 2024

TECHNICAL SKILLS

Languages: English, Spanish
Programming: Python (Pandas, NumPy, Scikit-learn, Jupyter), R, SQL (PostgreSQL)
Geospatial Analysis: ArcGIS, QGIS, GeoPandas, spatial data integration, remote sensing
Machine Learning: Gradient boosting, random forests, classification, feature engineering
Statistical Analysis: Multivariate analysis, predictive modeling, time series, data quality validation
Database & Infrastructure: Relational database design, ETL pipelines, cloud platforms (Azure), data workflows
Data Visualization: Tableau, Streamlit, scientific reporting, data documentation
Tools: Git, Jupyter Notebooks, Microsoft Office Suite, Excel (advanced)